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Robust machine-learning model for prediction of carbon dioxide adsorption on metal-organic frameworks

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ABSTRACT

Carbon dioxide (CO_2) capture is promising with economic and environmental benefits as an active climate-change mitigation approach. Nevertheless, the low adsorption capacity of traditional materials such as zeolites remains challenging. Metal-organic framework (MOF) has been proposed as a remedy, but its laboratory screening is costly and time-consuming. Hybrid machine learning and optimization models offer precise and rapid predictions of CO_2 uptake, effectively addressing the issue of inadequate hyperparameter tuning often seen in standalone machine learning models. In this study, hybrid machine-learning models are employed to accurately predict the CO_2 adsorption capacity of MOFs required for swift screening of MOF candidates, expediting the discovery of novel materials and fostering a more comprehensive understanding of CO_2 adsorption mechanisms. Six models, including multi-layer perceptron neural network (MLPNN) and least square support vector machine (LSSVM) algorithms, along with their hybrid counterparts utilizing particle swarm optimization (PSO) and growth optimization (GO) algorithms, are evaluated based on experimental data from previous studies. The experimental data consists of 475 data points and six input variables: Brunauer-Emmett-Teller specific surface area (SBET), pressure (P), temperature (T), Langmuir-specific surface area (SL), metal center (MC), and total pore volume (VT). Results demonstrate that the LSSVM-GO model outperforms the others, with an overall root mean squared error of 0.7484 and an R2 of 0.9798. The LSSVM-GO model's applicability domain is verified using a Williams plot, and a feature importance analysis conducted revealed that SBET has the greatest effect and VT has the least effect on its MOF CO_2-uptake predictions. The novelty of this study lies in developing a machine-learning-based approach that provides an accurate and cost-effective prediction of adsorption capacity for different MOF design experiments. This approach has the potential to significantly reduce screening costs and time required for various MOF designs, making it a more viable carbon capture strategy.

1. Introduction

Carbon dioxide (CO_2) plays a significant role in global warming, with atmospheric concentrations rising due to fossil fuel combustion in electricity generation, transportation, and industry [1–3]. Pachauri et al. [4] reported that CO_2 levels increased from 280 to 400 ppm, coinciding with a 0.8°C rise in global temperatures [2]. Predictions suggest that by the 22nd century, CO_2 concentrations could reach 600–700 ppm, potentially causing a 4.5–5°C increase in Earth's average temperature [5]. Many countries have introduced initiatives to reduce CO_2 emissions through efficient CO_2 capture strategies. These strategies aim to capture and sequester CO_2 (CCS), in appropriate storage media with options including subsurface reservoirs combined in some cases with enhanced oil recovery [6–9].

To date, various CO_2 capture and storage methods have been explored, including absorption techniques [10–12], membrane technologies [13,14] and carbon-based adsorbents [15,16], each showing a degree of promise. However, these methods face high operational costs, limited capacities, and complex regeneration requirements [17]. For a material to be deemed effective in the CO_2 capture process, it must exhibit superior CO_2 adsorption properties and allow for easy release of CO_2 during the regeneration phase. Enhancing the material's effectiveness involves structural modifications by incorporating a variety of

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Nomenclature

ALIGNN	Atomistic Line Graph Neural Network
ANN	Artificial neural network
AOC	Area over the curve
ARE	Average relative error
CIF	Crystallographic Information File
CO_2	Carbon dioxide
DL	Deep learning
ELM	Extreme learning machine
GCMC	Grand Canonical Monte Carlo
GO	Growth optimization
GP	Genetic programming
HML	Hybrid machine learning
hMOFs	Hypothetical metal-organic frameworks
LSSVM	Least square support vector machine
MC	Metal center
ML	Machine learning
MLP-BR	Multi-layer perceptron – Bayesian Regularization
MLP-LMA	Multi-layer perceptron-Levenberg Marquardt
MLPNN	Multi-layer perceptron neural network
MOFs	Metal-organic frameworks
OFI	Over-fitting index
P	Pressure
PSO	Particle swamp optimization
R^2	Coefficient of determination
RE	Relative error
REC	Regression error characteristics
RMSE	Root mean square root
RRMSE	Relative root-mean-square error
S_{BET}	Brunauer-Emmett-Teller (BET) specific surface area
S_L	Langmuir-specific surface area
T	Temperature
UMC	Unsaturated metallic centers
V_T	Total pore volume

functional groups and the tuning of molecules, which can significantly improve the efficiency of the CO_2-capturing process.

Metal-organic frameworks (MOFs) are crystalline materials with metal ions/ ion-clusters integrated with organic molecules to form rigid, porous structures of up to three dimensions. MOFs have been identified as promising materials for CCS, due to their adaptable nature, high-temperature stability, and easily adjustable chemical structures [18–21]. Li et al. [22] demonstrated how the active functional groups and MOFs' structure can be precisely controlled. MOF morphology and crystalline shapes influence their ability to withstand high temperatures [23–26]. Yeh et al. [24] demonstrated that MOFs synthesized from calcium and strontium exhibit remarkable thermal stability up to 450°C due to their micro-porosity. Modifying the functional groups on the pore walls of MOFs can endow them with characteristics suited to diverse applications [27,28].

3D MOFs are constructed from clusters of metal ions connected by organic molecule linkers, forming pores and channels that can adsorb and desorb molecules [29–31]. The choice of metal and linker affects the properties of MOFs. For example, IRMOF-1 contains zinc and terephthalic linkers, providing high-capacity adsorption pores. Some MOFs have unsaturated metallic centers (UMCs), offering more active sites for CO_2 and enhancing the bond between the MOF structure and CO_2 [32]. The versatility and unique properties of MOFs makes them promising for carbon capture and storage (CCS) applications.

MOFs are favored over traditional materials like zeolites for CO_2 adsorption, due to their larger pore sizes, enabling better molecular diffusion within and across crystal structures [33–36]. This advantage, combined with their tunable porosity and customizable surface chemistry, positions MOFs as attractive candidates for CO_2 adsorption [37–39]. However, the experimental assessment of MOFs' adsorption capacity is time-consuming and costly, often requiring adaptation of specific isotherms [40].

Comprehensive understanding of CO_2 adsorption mechanisms on MOFs is essential for the rational design and optimization of these materials. CO_2 adsorption in MOFs is influenced by various factors, including pore size [36], surface area [36], functional groups [37], and the presence of open metal sites [37]. The rapid screening of metal-organic frameworks (MOFs) for CO_2 adsorption is a critical step in accelerating the discovery of effective materials for carbon capture applications. Traditional experimental methods for evaluating the performance of MOFs are labor-intensive and time-consuming, often requiring extensive synthesis and characterization efforts.

Machine learning (ML) offers a powerful alternative by enabling high-throughput computational screening of MOF libraries, thus expediting the identification of promising candidates. For example, a study by Mahajan and Lahtinen [37] reveals that the presence of open metal sites and specific pore geometries are the most critical factors influencing CO_2 adsorption in MOFs. These insights enable the targeted design of new MOFs with enhanced adsorption properties. In addition to structural features, the dynamics of CO_2 adsorption and desorption processes are crucial for practical applications. Kinetic studies, such as those conducted by Gibson et al. [36], provide insights into the rates of CO_2 uptake and release, which are important for evaluating the performance of MOFs in cyclic adsorption processes. Understanding these kinetics helps in designing materials that can rapidly adsorb and desorb CO_2, making them suitable for industrial applications [38]. Thus, fostering a comprehensive understanding of CO_2 adsorption mechanisms on MOFs requires a multidisciplinary approach that combines machine learning, computational modeling, and experimental validation. By integrating these methods, researchers can gain deep insights into the factors influencing CO_2 adsorption, guiding the rational design and optimization of MOFs for carbon capture applications.

The advantages of cost and time-efficiency offered by machine learning (ML) without relying on oversimplified assumptions is an alternative for CO_2 uptake predictions. Machine learning techniques known for capturing complex non-linear relationships, are increasingly used in predictive modeling [41–45]. ML algorithms such as support vector machines, radial basis networks, fuzzy logic, and colony optimization have found applications in various industries, including carbon capture [46–50], demonstrating their effectiveness in enhancing predictive accuracy. These studies use ML to predict the non-linear adsorption characteristics of CO_2 by MOFs.

Various researchers have used ML to predict CO_2 uptake in different metal-organic frameworks (MOFs). Various descriptors, such as geometrical, chemical, topological, and energy-based, can model CO_2 uptake capabilities. Geometrical descriptors, which focus on the pore environment in MOFs, are commonly used in ML to predict gas adsorption and storage capacities. These descriptors include metrics like volumetric surface area, gravimetric surface area, pore volume, density, and void fraction, providing a general overview of the material's properties. They are generally good indicators of gas storage capabilities in porous materials, making them suitable for ML applications [51]. Choudhary et al. [50] used the Atomistic Line Graph Neural Network (ALIGNN) to predict CO_2 adsorption in MOFs, utilizing the hypothetical MOF (hMOF) database and grand canonical Monte Carlo (GCMC) simulation-based isotherms. However, GCMC simulations require significant computational resources, making them time-consuming and costly when working with large datasets containing millions of MOFs.

Additionally, these simulations need detailed information about the characteristics of the MOFs.

Lu et al. [51] created a deep-learning model to predict CO_2 working capacity in metal-organic frameworks (MOFs) at low pressure, using only Crystallographic Information File (CIF) data instead of experimental results. This approach involves transforming 3D CIF data into 2D representations, which can lead to errors and uncertainties. To reduce issues and the computational costs related to energy-based and topological descriptors, some researchers have combined geometrical descriptors with experimental conditions for modeling the CO_2 absorption in MOFs. Nait Amar et al. [52] investigated CO_2 uptake predictions in MOFs with four models: genetic programming (GP), extreme learning machine (ELM), and multi-layer perceptron (MLP) with Bayesian Regularization (MLP-BR) or Levenberg-Marquardt (MLP-LMA) algorithms. This study did not account for the impact of different metal centers in MOFs. Similarly, Abdi et al. [53] used ensemble-decision-tree techniques, including Random Forest (RF), Light Gradient Boosting Machine (LightGBM), Extreme Gradient Boosting (XGBoost), and Categorical Boosting (CatBoost), to predict CO_2 adsorption in 19 MOF variants. They used four inputs: pressure, temperature, pore volume, and specific surface area of the MOFs. Sensitivity analysis revealed that temperature had a negative impact on CO_2 adsorption, while specific surface area and pressure were the most significant factors. However, this study also did not consider the type of metal centers in their models.

These studies demonstrate the potential of ML models to predict CO_2 uptake in MOFs and optimize carbon capture processes. The literature review identifies the ML approaches previously applied to predict CO_2 adsorption capacity in MOF systems. These standalone ML models have

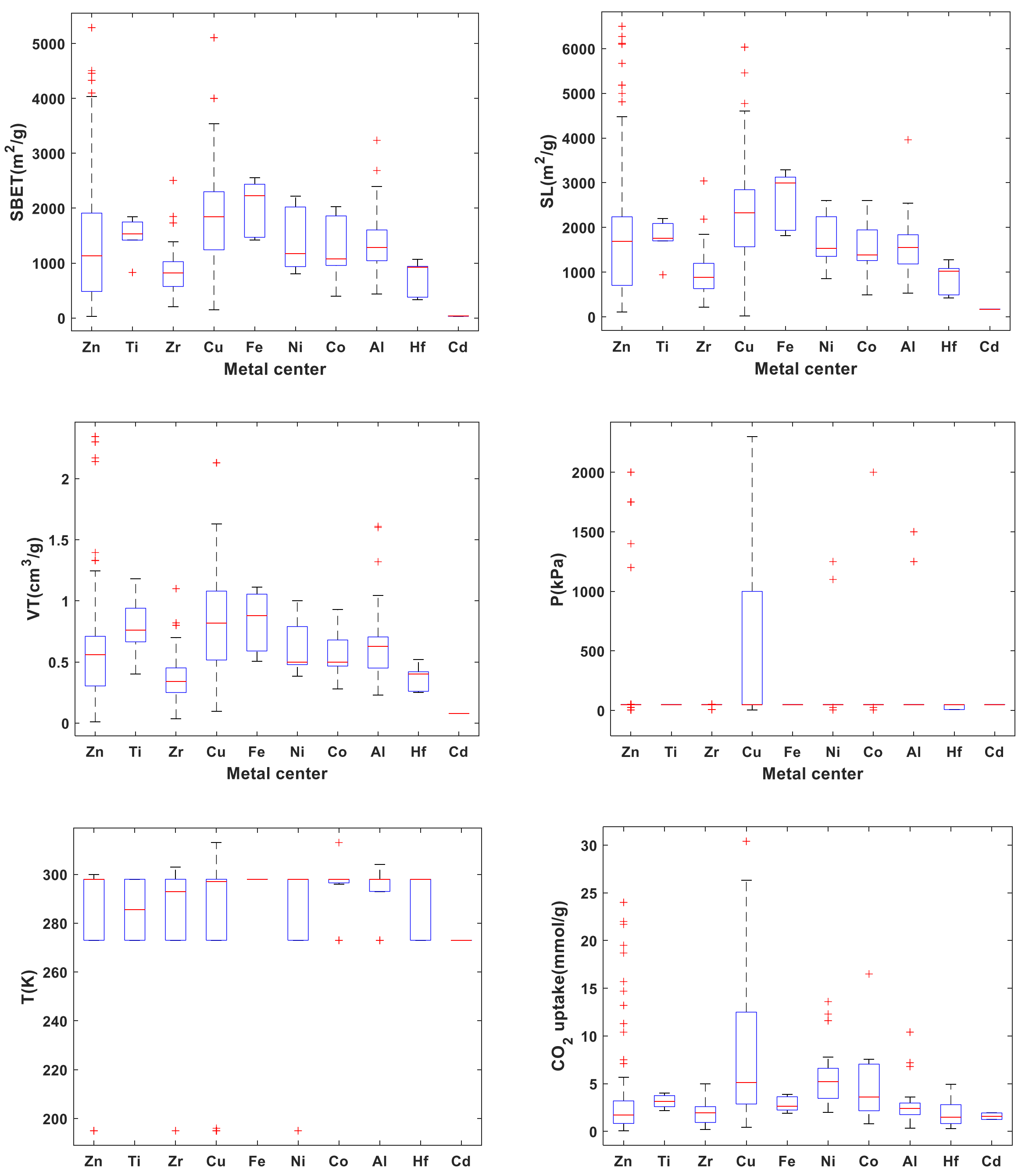


Fig. 1. Box-and-whisker plots display outliers for each variable before normalization (Plus signs represent points outside the whiskers).

several drawbacks such as poor hyperparameter tuning and slow convergence. To enhance the predictive accuracy of traditional machine learning approaches, researchers often integrate optimization algorithms into their models. Little attention has been given to using hybrid machine learning (HML) models for robust prediction of CO_2 uptake in MOFs systems, and the capabilities of HML models in this regard have yet to be evaluated. LSSVM and MLPNN hybridized with Growth optimization (GO) and particle swamp optimization (PSO) models are innovative approaches that have not previously been adapted for predicting CO_2 uptake in MOFs system. These HML models have shown high accuracy and reliability in other fields of engineering and scientific applications [54,55], and are therefore selected for his study.

This paper is organized into six sections. Section 2 outlines the dataset collection and processing methods. Section 3 discusses research methodologies. Sections 4 and 5 present the study's findings, along with analysis and discussion. Section 6 highlights the study's limitations and provides suggestions for future research. Section 7 concludes with a summary of the key findings and their significance.

2. Data collection and characterization

A comprehensive experimental database was compiled to predict CO_2 uptake in metal-organic frameworks (MOFs) using HML algorithms, using supplementary materials from Li et al. [56]. This study utilized data from various literature sources, accumulating 475 data points representing different MOFs with diverse metal centers. The aim was to improve the predictive accuracy reported by Li et al. [56]. After data pre-processing, six key attributes were identified as predictors for CO_2 uptake in MOFs, encompassing physical and operational parameters: Brunauer-Emmett-Teller (BET) specific surface area (S_{BET}, in m^2/g), Langmuir specific surface area (S_L, in m^2/g), total pore volume (V_T, in cm^3/g), pressure (in kPa), temperature (in K), and the metal composition forming the MOF's metal center. The target variable was CO_2 uptake capacity (in mmol/g). Fig. 1 provides a box-and-whisker plot illustrating the distribution of values for each numerical feature in the compiled MOFs-CO_2-adsorption database, grouped by the MOF metal center.

This dataset contains data points across various metal-organic framework (MOF) metal centers, with the following distribution: 167 points for Zn, 12 for Ti, 65 for Zr, 106 for Cu, 10 for Fe, 32 for Ni, 19 for Co, 44 for Al, 30 for Hf, and 2 for Cd. Fig. 1 reveals that the data from Zn and Cu centers exhibit a wider range due to the larger number of data points. On the other hand, Fe centers have notably higher median values for SBET, SL, and VT, but lower median CO_2 uptake compared to other metal centers. This discrepancy between CO_2 adsorption and surface area might be due to uncertainties in N_2-based surface area analysis [57]. Samples with Cd centers have the lowest feature values, possibly because of their low pore diameter, which likely limit their CO_2 adsorption, as the pore diameter needs to be larger than the kinetic diameter of CO_2 [57]. Further research is suggested to explore Cd-containing MOFs under high temperature and pressure conditions. Table 1 provides a statistical summary of the dataset, which includes MOFs with ten distinct metal centers: aluminum, nickel, zinc, zirconium, titanium, copper, cadmium, cobalt, iron, and hafnium.

Table 1
Statistical summary of the variables comprising the CO_2 uptake compiled 475-point dataset.

Variables	S_{BET} (m^2/g)	S_L(m^2/g)	V_T (cm^3/g)	P(kPa)	T(K)	CO_2 Uptake (mmol/g)
Mean	1457.42	1747.7	0.645	363.5	287.15	4.3065
Std	1011.54	1258.9	0.460	867.4	19.73	5.4534
Min	30.00	19.0	0.012	10.0	195.00	0.0800
25 %	809.75	978.8	0.380	100.0	273.00	1.2947
50 %	1218.00	1571.0	0.520	100.0	298.00	2.4400
75 %	1911.09	2234.5	0.795	100.0	298.00	4.3661
Max	5290.00	6500.0	2.345	4600.0	313.00	30.4000

3. Methodology

A flowchart (Fig. 2) displays the sequence of processing steps involved in the applied workflow.

To develop predictive models for CO_2 uptake using six input variables (S_{BET}, S_L, V_T, P, T, MC), the data is first filtered to remove outliers. The dataset is then split into training and testing subsets to assess model performance. Predictive algorithms LSSVM and MLPNN, combined PSO and GO optimizers, are used to build accurate and generalizable models. The structure of MLPNN and the kernel function for LSSVM are optimized with the training data. Hyperparameter tuning is performed to enhance the performance of the predictive algorithms. After building the models, they are tested with unseen data and evaluated using error metrics and the coefficient of determination. Many detailed explanations of these ML and optimizer algorithms are available in literature and are summarized in the Supplementary File.

To ensure robust model selection, overfitting evaluation, scoring, and regression error characteristics (REC) are used to determine the most predictive and generalizable model. Additionally, William's plot helps identify outliers, while Shapley additive explanation (SHAP) and partial dependence plots (PDP) offer insights into the best model's interpretability and the influence of individual input variables on the output.

The following subsections describe the ML/optimizer hybridization workflow applied to create predictive models of CO_2 uptake, and the criteria with which their prediction performance is evaluated.

3.1. Development of hybrid deep learning predictive model of CO_2 uptake

Optimizing ML algorithm's hyperparameter values is essential for achieving precise and generalizable models [58,59]. This study uses PSO and GO algorithms to fine-tune the hyperparameters of LSSVM and MLPNN models. Combining optimization and predictive algorithms requires that the optimizers consider all the decision variables, which corresponds to the hyperparameters in each predictive algorithm. In MLPNN, these decision variables relate to the network structure, while in LSSVM, they relate to the choice of kernel function. Selecting the appropriate structure for MLPNN and the most effective kernel function for LSSVM is dependent on the characteristics of the studied dataset.

PSO is initially applied to determine the number of MLPNN hidden layers and neurons in each layer. Then PSO and GO are separately applied to determine the optimal MLPNN weights and biases. An error metric, typically RMSE, is computed to represent the cost of the selected hyperparameters, providing feedback to the optimization algorithms. In subsequent iterations, the optimizers refine their solutions by adjusting within the available search spaces. This iterative process continues for a specified number of iterations (Fig. 3).

In LSSVM, selecting the appropriate kernel function depends on the problem's conditions and often requires a trial-and-error approach. After choosing the kernel function and specifying the number of hyperparameters, optimization algorithms start by initializing random values for these hyperparameters in the search space during the first iteration. These values are used to build the LSSVM model to predict the target parameter. The error from this prediction is then calculated and used as feedback for the optimization algorithms, representing the cost of the chosen hyperparameters. In subsequent iterations, the optimizers adjust the hyperparameters to improve the solution, aiming to reduce the prediction error. This iterative process continues for a specified number of iterations (Fig. 4).

3.2. Developed model's evaluation criteria

Multiple standards are used to evaluate the effectiveness of

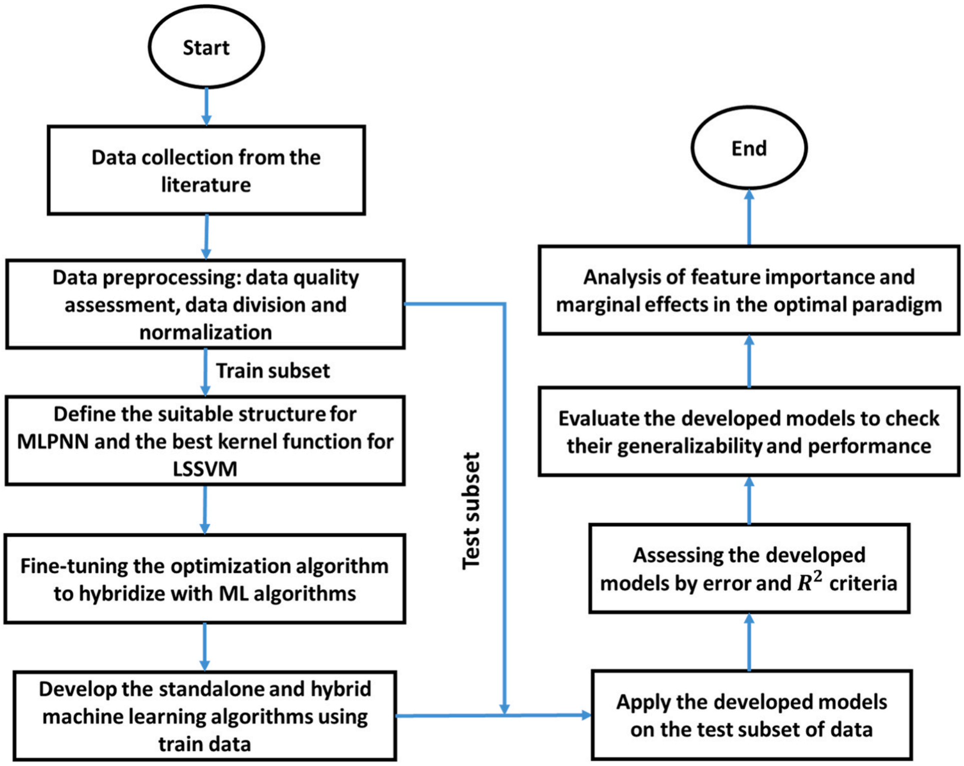


Fig. 2. Summary of the process steps constituting the applied workflow.

predictive models, focusing on minimizing errors and ensuring optimal generalizability. Each metric has unique strengths and weaknesses, so several are used in the assessment process to balance these factors when choosing the best model. Six performance metrics are examined for both standalone and hybrid machine learning (HML) models.

Computing error (Err) values involve comparing experimental and predicted CO_2 uptake values for each data point using Eq. (1). The division of Err values by measured CO_2 uptake values yields the relative error (RE) (Eq. (2)). Subsequently, the average relative error (ARE) is determined by aggregating total RE values and the total number of data points (Z) (Eq. (3)). The root mean square error (RMSE), a pivotal prediction-error metric, is computed using Eq. (4). Gandomi et al. [60] introduced the performance index (PI), which concurrently considers the correlation coefficient (R) and relative RMSE (RRMSE) as calculated with Eq. (5). The RRMSE value is derived by dividing the RMSE value by the average measured CO_2 uptake value using Eq. (6). These prediction-performance metrics mentioned above primarily measure errors, where lower values indicate superior prediction performance. Conversely, the coefficient of determination (R^2) (Eq. (7)) is regarded as a positive criterion, with higher values signaling enhanced prediction performance. The a20-Index, another positive prediction-performance metric [61,62], is determined by dividing the number of points situated between the Y (where Y = predicted CO_2 uptake) = 0.8X (where X = measured CO_2 uptake) and Y = 1.2X lines (m20) by the total number of data points (Eq. (8)). A higher value of the a20-Index signifies a greater concentration of data points around the Y = X line.

$$Err_i = x_{i,experiment} - x_{i,predicted} \tag{1}$$

$$RE_i = \frac{Err_i}{x_{i,experiment}} \times 100 \tag{2}$$

$$ARE = \frac{\sum_{i=1}^{Z} RE_i}{Z} \tag{3}$$

$$RMSE = \sqrt{\frac{1}{Z}\sum_{i=1}^{Z} Err_i^2} \tag{4}$$

$$PI = \frac{RRMSE}{1+r} \tag{5}$$

$$RRMSE = \frac{RMSE}{\overline{x_{experiment}}} \tag{6}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{Z} Err_i{}^2}{\sum_{i=1}^{Z}\left(x_{i,predicted} - \frac{\sum_{i=1}^{Z} Err_i^2}{Z}\right)^2} \tag{7}$$

$$a20 - Index = \frac{m20}{Z} \tag{8}$$

4. Results

4.1. Data characterization

Understanding how the dependent and independent variables correlate is essential in ML modeling. A Pearson correlation analysis was conducted to evaluate the correlation between the variables considered and expressed as a heat map (Fig. 5).

Most features show a direct correlation with CO_2 uptake, except temperature (T), which has a weak inverse correlation, consistent with experimental studies [63]. As temperature increases, CO_2 uptake decreases. Pressure (P) has the strongest linear correlation with CO_2 uptake, due to the role it plays in pore-filling dynamics and pressure-dependent adsorption mechanisms in MOFs. MOFs' high surface area and porosity allow significant CO_2 adsorption, with the process

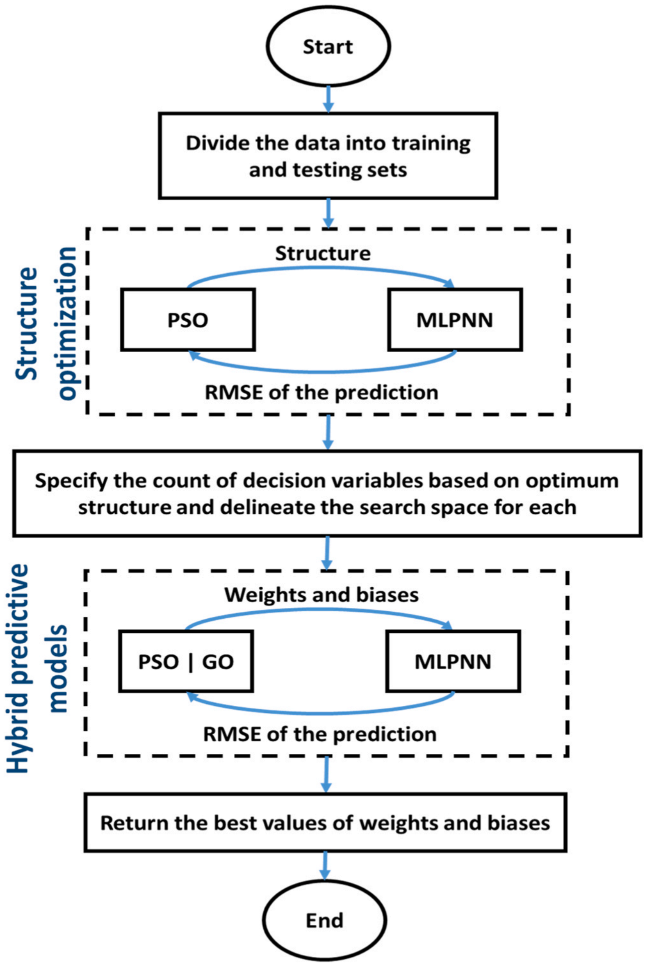


Fig. 3. Flowchart for the development of hybrid prediction and optimization algorithms to identify the optimal structure of the MLPNN model and its optimal weights and bias values.

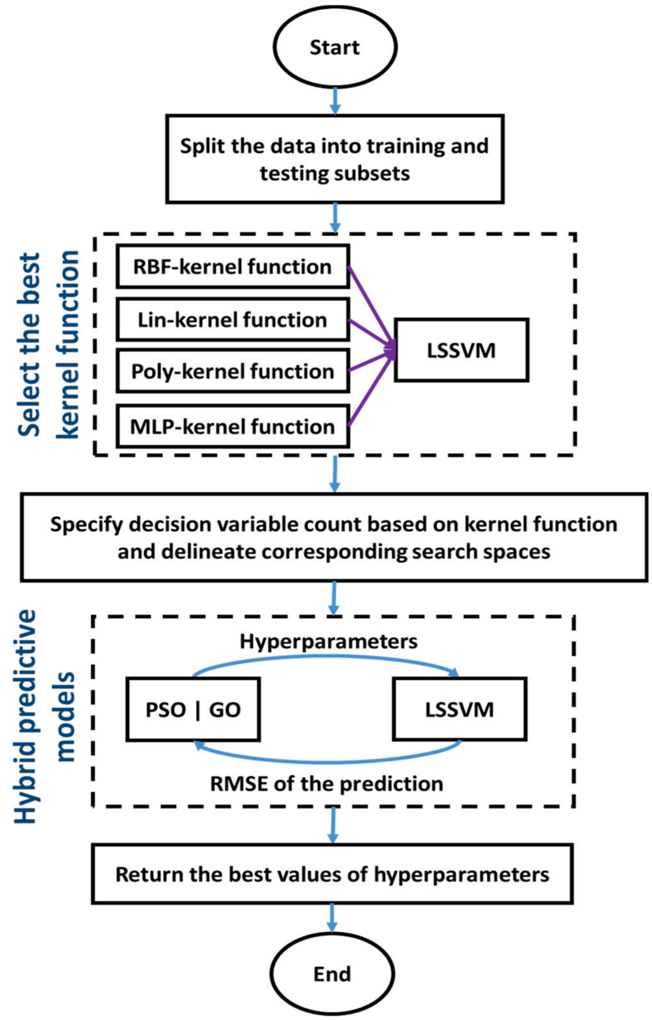


Fig. 4. Flowchart for the development of hybrid prediction and optimization algorithms to identify optimal hyperparameter values for the LSSVM model.

being pressure dependent. At lower pressures, CO_2 uptake is driven by physisorption, where CO_2 is held by weak van der Waals forces. As pressure rises, more CO_2 molecules fill the MOF pores, although further increases in pressure have minimal effect once the pores are saturated. At higher pressures, chemisorption, involving stronger chemical interactions might occur.

Additionally, the weakest correlation is with metal center (MC), as demonstrated by previous studies [56]. Strong Pearson's correlation coefficients among S_{BET}, S_L, and V_T features suggest a degree of interdependence. These findings highlight the complex dynamics affecting CO_2 uptake in MOFs. Understanding these correlations can inform the design of MOF materials optimized for CO_2 capture under varying pressure conditions.

4.2. Pre-processing

After evaluating the data and removing 12 incomplete data points, the remaining 475 data points were divided into training and testing categories with varying separation ratios from 0.6 to 0.95, as determined by the PSO algorithm. The LSSVM model with an RBF kernel was applied to the training set, and its prediction error was used as feedback to the PSO algorithm to optimize the separation ratio. Fig. 6 illustrates the reduction in error over several iterations of the PSO algorithm, with no significant reduction after approximately 20 repetitions, indicating sufficient repetitions. The optimization results showed that the optimal separation ratio was 82 % for training (390 data points) and 18 % for

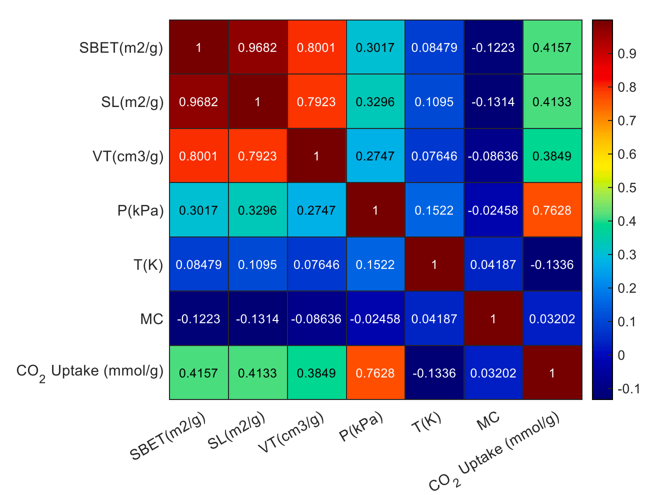


Fig. 5. Pearson's correlation coefficient heat map of the input and target parameters of the compiled dataset.

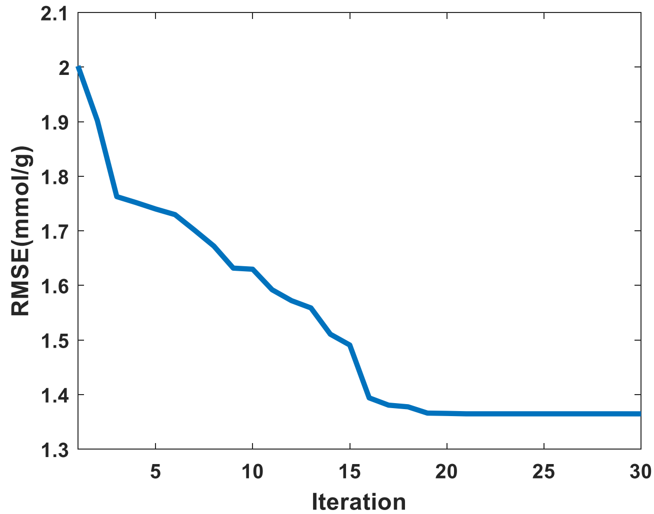


Fig. 6. LSSVM prediction-error fluctuations for the test data across thirty iterations of the PSO algorithm, aimed at optimizing the separation ratio between the training and test data.

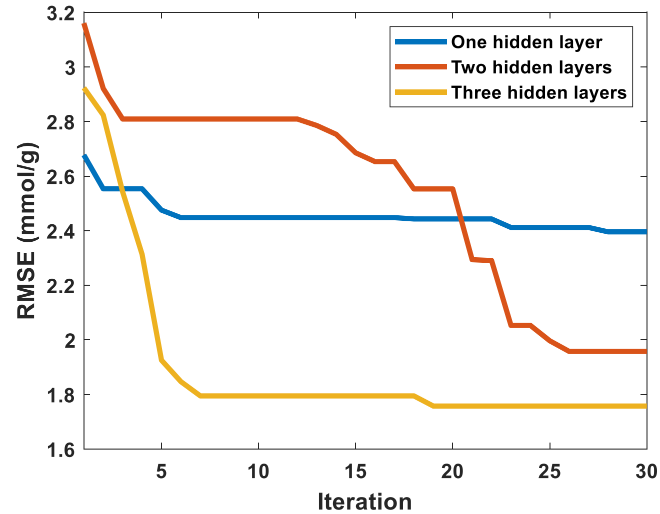


Fig. 8. RMSE changes in different iterations of the PSO algorithm to find the optimal number of neurons and learning rate in MLPNN model configured with one, two, and three hidden layers.

testing (85 data points). This configuration demonstrated the LSSVM model's high accuracy and generalization capability.

4.3. Tuning the machine learning algorithms

The ideal kernel function for LSSVM and the optimal structure for MLPNN were selected according to problem-specific conditions. To identify the best LSSVM model, various kernel functions listed in Table S1 were applied to the training data. Fig. 7 compares these models using the RMSE metric.

The LSSVM model with the Radial Basis Function (RBF) kernel achieved the lowest RMSE of 1.5256 mmol/g and was chosen for further development. The RBF kernel has two key hyperparameters: regularization (γ) and the RBF kernel function parameter (σ^2). γ optimizes model performance on training data while controlling model complexity—a higher γ indicates less regularization, leading to a more complex, non-linear model. The σ^2 determines how many neighboring data points influence the model—a higher σ^2 includes more neighboring data points, resulting in a more non-linear model. Optimization algorithms aim to find the best combination of γ and σ^2 by minimizing the cost function.

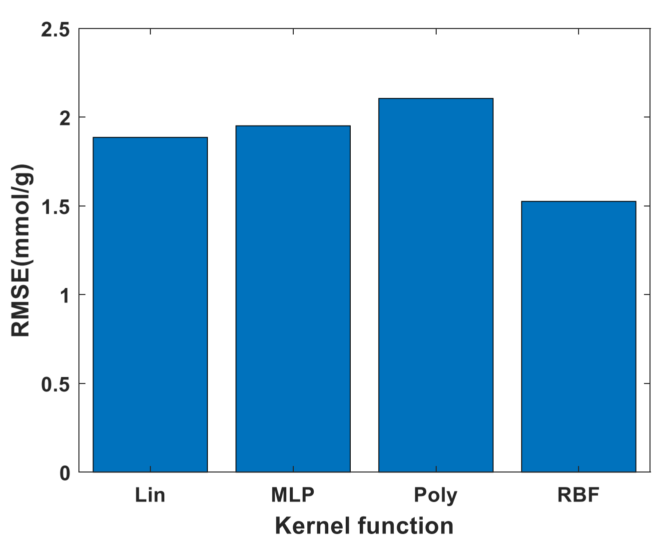


Fig. 7. Comparison of LSSVM model's performance applying four different kernel functions for predicting CO_2 uptake using training data.

Both LSSVM and MLPNN algorithms were optimized using PSO and GO algorithms. Fig. 8 shows the error fluctuations across various iterations of the PSO algorithm during the development of CO_2 uptake prediction models using MLPNN with different layer structures. The MLPNN model with three hidden layers, each with its optimized number of neurons, yielded lower error values compared to models with fewer layers.

Table 3 details the recommended number of neurons for each hidden layer in different MLPNN structures. For the three-layer structure, the optimal configuration was 9, 17, and 9 neurons for the first, second, and third layers, respectively. The learning rate was also optimized alongside the number of neurons in each layer.

The optimal learning rate values for different structures are also displayed in Table 2. Since the three-layer structure is chosen for developing CO_2 uptake prediction models, a learning rate of 0.8267 was utilized for the optimum MLPNN model.

Table 3 shows the results from sensitivity analysis, revealing optimal optimizer control parameter values. These optimized settings were applied in the algorithmic analysis of the dataset.

4.4. Developing predictive ML and HML models

Fig. 9 illustrates the trend of reducing RMSE across various iterations of the optimization algorithms, applied to find the optimal hyperparameters for LSSVM and MLPNN using the training dataset. The lowest error rate was achieved with the LSSVM-GO hybrid algorithm, indicating that for the training data, LSSVM outperformed MLPNN, and GO outperformed PSO, when applied to both ML models.

Fig. 10 shows cross plots of measured and predicted CO_2 uptake values on training data for each predictive algorithm. At high CO_2 uptake values, there is a noticeable deviation from the Y = X line in the MLPNN, MLPNN-PSO, and LSSVM-PSO models. This deviation is most pronounced in the simple MLPNN model. Specifically, the MLPNN-PSO

Table 2
The optimal number of neurons in the hidden layers and the optimal learning rates for MLPNN when single, double, or triple hidden layers are used.

Controllable parameters	Number of hidden layers		
	1	2	3
Optimal number of neurons in each layer	12	6 and 20	9, 19, and 9
Learning rate	0.8062	0.8295	0.8267

Table 3
Adopted values for the control parameters of the PSO and GO algorithms hybridized with predictive algorithms.

Parameters	MLPNN		LSSVM	
	PSO	GO	PSO	GO
Maximum iteration	100	100	100	100
Population size	80	70	70	50
w	0.96	NA	0.97	NA
C1	2.05	NA	2.05	NA
C2	2.05	NA	2.05	NA
Retention probability	NA	0.001	NA	0.001
Reflection probability	NA	0.3	NA	0.3
Number of upper-level individuals	NA	5	NA	5

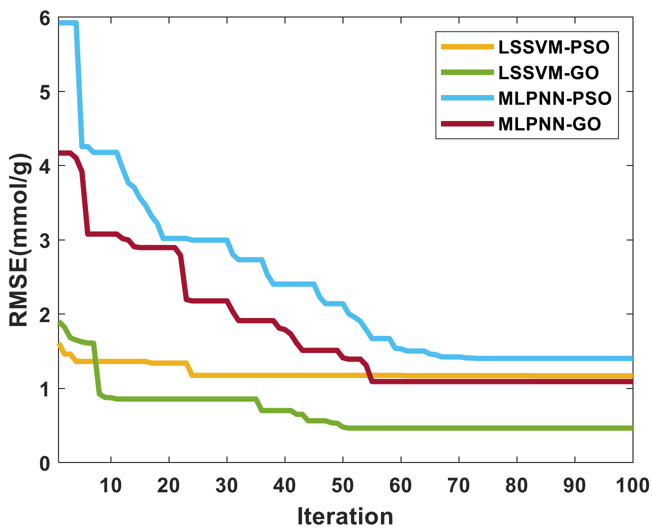


Fig. 9. Variations of RMSE for different iterations of the hybrid algorithms for developing CO_2-uptake predictive models using the training data.

and LSSVM-PSO models tend to underestimate, while the simple MLPNN model tends to overestimate. However, the LSSVM-GO model does not exhibit this deviation, demonstrating superior accuracy across the entire CO_2 uptake range. This indicates that the GO algorithm is better than the PSO algorithm at avoiding prediction errors at high CO_2 uptake values.

Table 4 compares CO_2 uptake prediction models using simple and hybrid algorithms, based on various error metrics and R^2 values from the training data. The hybrid models, particularly those using the GO algorithm, achieve significantly lower RMSE than the simple ML models, indicating that the optimizers positively impact performance. All six-performance metrics confirm that the LSSVM-GO model outperforms other models with the training dataset.

Fig. 11 cross plots the measured and predicted CO_2-uptake values for each of the trained models applied to the test data. As was the case with the training data, only the LSSVM-GO model's predictions track the Y = X line. This deviation phenomenon is more pronounced in the simple and hybrid MLPNN models compared to the LSSVM model, confirming the superior performance of the LSSVM algorithm with this dataset.

Since only a small portion of the data collected has high uptake values, explanations are provided in Appendix A to outline the distinct conditions associated with these higher uptake values compared to those with lower values.

Table 5 compares the prediction performance, in terms of six metrics of the trained CO_2-uptake prediction models applied to the test data. The LSSVM-GO model's performance substantially surpasses that of other models evaluated. The optimized control values for the LSSVM's RBF kernel are $\gamma = 277.6672$ and $\sigma^2 = 0.0488$.

5. Discussion

In this section, the efficacy of the developed models is further assessed.

5.1. Overfitting index (OFI) assessment of the models

Overfitting can be problematic for ML and HML models [64]. An overfitting assessment is therefore required for each model. Various techniques are available to do that [65–67] but the overfitting index [68] (OFI; Eq. (9) is used here.

$$OFI = \left(\frac{Z_{tr} - Z_{ts}}{Z}\right)PI_{tr} + 2\left(\frac{Z_{ts}}{Z}\right)PI_{ts} \tag{9}$$

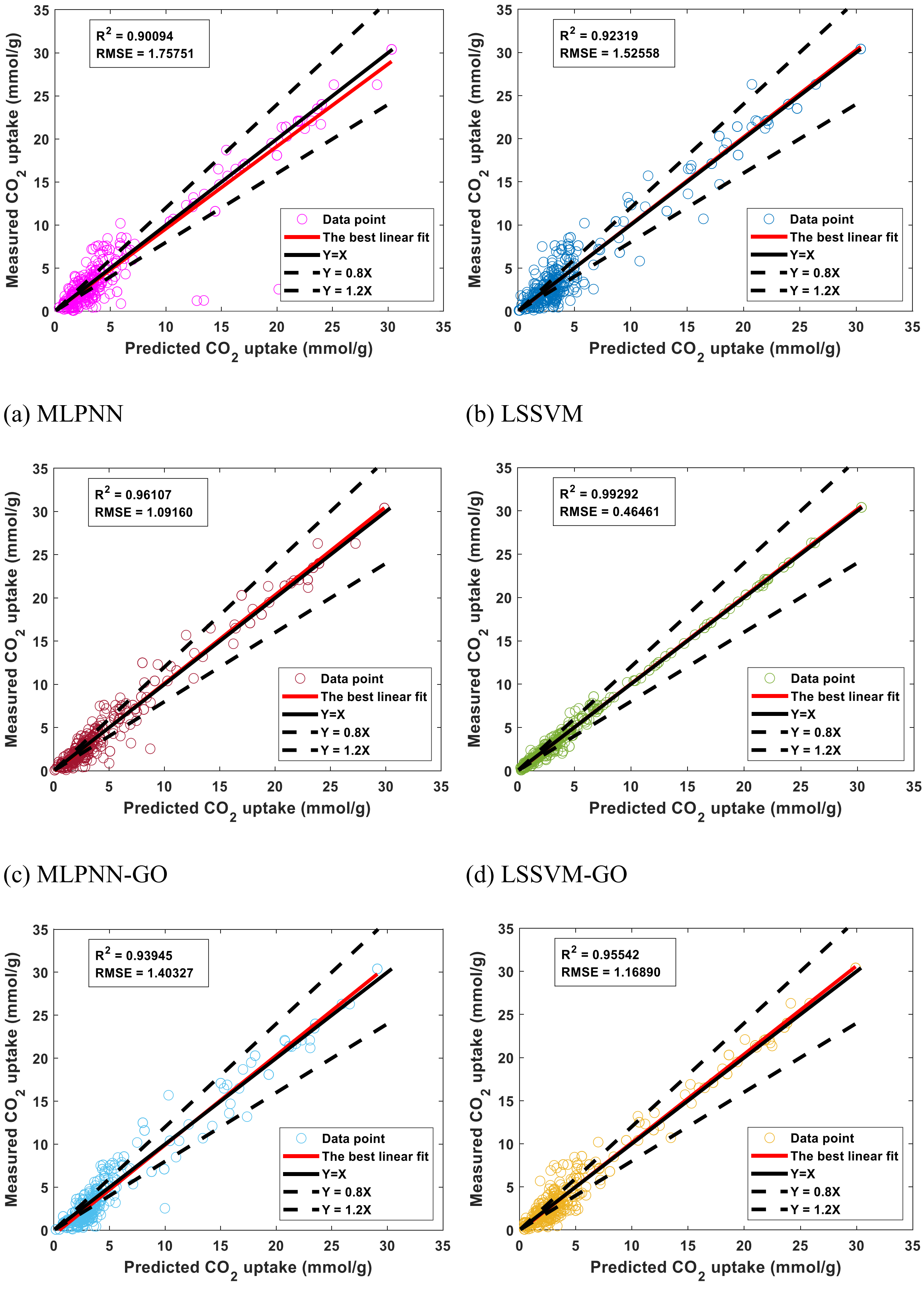
Where PI (Prediction Index; Eq. (5)) of training (PI_{tr}) and testing (PI_{ts}) results are adjusted by the relative number of data points (Z_{tr} for training and Z_{ts} for testing).

Table 6 presents the calculated OFI values for the developed models ranking them LSSVM-GO (best) >MLPNN-GO > LLSVM-PSO > MLPNN-PSO > LSSVM > MLPNN (worst) and highlighting that the trained LSSVM-GO model is more generalizable and less prone to overfitting than the other models. This performance ranking of the predictors is further confirmed by score and regression error characteristic analysis (see Supplementary File).

5.2. Model performance under consistent temperature and pressure conditions

Analysis of uptake values revealed that lower uptake values correspond to low pressure, while higher uptake values align with high pressure. Additionally, the distribution of data points is more concentrated in the lower uptake range than in the higher range. Given that most data points are clustered around similar (low) pressure and temperature conditions, it is suggested that the LSSVM-GO model effectively predicts uptake for two different MCs. In other words, the model has successfully captured the influence of MC type, enabling it to differentiate uptake levels for different MCs under similar temperature and pressure conditions. To assess the model's ability to distinguish uptake performance for two different MOF materials under the same T, P conditions, gathered data were grouped into three clusters using the K-means method. This clustering approach allowed data points with similar temperature and pressure conditions to be organized together, enabling a focused comparison of the model's predictive accuracy for uptake within each cluster. Through this organization, an evaluation was made of how effectively the model captures the distinct absorption patterns of different MOF materials under identical experimental conditions. Table 7 shows the minimum and maximum values of pressure and temperature in each of the specified clusters, with most of the data are located in cluster 1. It is also evident that the clusters overlap in terms of temperature values, yet they differ significantly when it comes to pressure values. Fig. 12 illustrates the clustering results, with data points color-coded by cluster. Many data points share the same T, P values, resulting in some overlap, which might visually suggest fewer data points than are actually assessed.

To evaluate the model's predictive performance of the model generating the least errors (LSSVM-GO) for MOFs with different MCs under similar T, P conditions, a comparison was made between the predicted CO_2 uptake values using LSSVM-GO model and the measured ones, grouped by cluster. Fig. 13 illustrates the distribution of data points within each cluster for both the CO_2 uptake predictions generated by the LSSVM-GO model and the measured values of CO_2 uptake, categorized by the MC type in MOFs considered. In Cluster 1, the distribution pattern of the measured values of CO_2 uptake (Fig. 13a) is close to that of predicted CO_2 uptake values (Fig. 13b). Minor discrepancies are observed due to prediction errors, but the overall resemblance



(a) MLPNN (b) LSSVM

(c) MLPNN-GO (d) LSSVM-GO

(e) MLPNN-PSO (f) LSSVM-PSO

Fig. 10. Cross plots comparing measured and predicted CO_2 uptake utilizing (a) MLPNN, (b) LSSVM, (c) MLPNN-GO, (d) LSSVM-GO, (e) MLPNN-PSO, and (d) LSSVM-PSO models on the training data.

Table 4
Comparison of the prediction performances for simple and hybrid models applied to the CO_2 uptake training subset.

Type	Models	ARE	RMSE (mmol/g)	RRMSE	R^2	PI	a20-index
Simple	LSSVM	0.3659	1.5256	0.3414	0.9232	0.1741	0.3641
	MLPNN	0.6184	1.7575	0.3932	0.9009	0.2018	0.4462
Hybrid	**LSSVM-GO**	**0.0832**	**0.4652**	**0.1040**	**0.9929**	**0.0521**	**0.7949**
	MLPNN-GO	0.2087	1.0916	0.2442	0.9611	0.1233	0.5513
	LSSVM-PSO	0.3052	1.1689	0.2615	0.9554	0.1324	0.4590
	MLPNN-PSO	0.6085	1.4033	0.3140	0.9394	0.1594	0.3744

highlights the model's strong performance in predicting CO_2 uptake values under similar temperature and pressure conditions across considered MOFs with different MCs. This indicates that the model effectively captures the influence of MC type in MOFs on the CO_2 uptake, even under similar environmental conditions.

In Cluster 2, the comparison between measured (Fig. 13c) and predicted (Fig. 13d) CO_2 uptake values show complete alignment in the distribution of values under similar temperature and pressure conditions. This further reinforces the model's accuracy in predicting CO_2 uptake for the MOFs with different MCs under similar conditions. Similarly, in Cluster 3, the measured (Fig. 13e) and predicted (Fig. 13f) uptake values exhibit perfect similarity, emphasizing the model's ability to differentiate the effects of MC type in MOFs when predicting CO_2 uptake. Thus, it can be concluded that the LSSVM-GO model is highly effective in predicting CO_2 uptake values for MOFs with different types of MC under similar temperature and pressure conditions.

5.3. *Williams plot analysis of the CO_2 uptake best prediction model*

The William plot cross plots standardized residuals and leverage values from the Hat matrix (Fig. 14), helping to identify prediction outliers with high residuals and those with significant leverage on the dataset [69,70]. The Hat matrix (H) is derived from the two-dimensional dataset (number of variables (m) versus number of data points (n)), as detailed in Eq. (10) [71].

$$H = X\left(X^T X\right)^{-1} \cdot X^T \tag{10}$$

The diagonal elements of the matrix described in Eq. (10) represent leverage across n dimensions. An upper leverage limit (H*) can be calculated based on the m and n. Values exceeding that threshold identify data points warranting closer scrutiny in terms of the influence they have on model predictions [71].

Fig. 14 illustrates the data used by the LSSVM-GO model for predicting CO_2 uptake across the full dataset. About 98.74 % of the data have leverage values between 0 and 0.0758, the upper limit (H*), with only 6 data points exceeding this threshold. Additionally, 18 records fall outside the standardized residual range of −3–3, identified as "suspected outliers." Overall, approximately 94.95 % of the data lies within the model's applicable domain, indicating that only 5.05 % (24 out of 475 records) could be outliers. William's plot analysis suggests that the LSSVM-GO model is not heavily impacted by outliers, given the small number of records that deviate significantly. Thus, the outlier and out-of-leverage data points are unlikely to substantially affect the model's predictions.

5.4. *Feature importance analysis*

Assessing the impact of input features on predictive models is crucial for building effective ML/HML models. It helps understand the relationships between input and output variables, allowing analysts to identify key features that significantly affect the model's performance. This process facilitates adjustments to improve prediction accuracy and supports better decision-making. The SHAP (SHapley Additive exPlanations) tool was used to visualize the relative contributions of individual features to the model's predictions.

Fig. 15 illustrates the detailed impacts of features and summarizes their importance in the LSSVM-GO model's predictions using SHAP. In Fig. 15a, the P feature shows a polarity shift with negative SHAP values at lower levels and positive values at higher levels, indicating variability in its impact. Similarly, the SL feature trends with negative SHAP values at low levels and positive SHAP values at high levels. In contrast, the SBET attribute shows no clear trend.

The T characteristic displays a consistent pattern, with higher values corresponding to low-to-positive SHAP values. SHAP values for the MC and V_T features are close to zero, indicating a minimal impact on the model. According to the SHAP summary plot (Fig. 15b), S_{BET}, P, and T have the most significant influence on CO_2-uptake predictions, while MC, SL, and VT have the least impact. Although SBET has a high Pearson's correlation coefficient (involving parametric / linear assumptions) with SL, it has less effect on the output of the LSSVM-GO model. This discrepancy can be due to two factors. First, while SBET and SL are indeed highly correlated, they reflect different surface characteristics because of the way they are measured: SBET accounts for both microporous and mesoporous regions, whereas SL primarily represents microporous areas. This difference implies that, under the experimental conditions in this study, the mesoporous contribution (captured by SBET but not SL) might significantly influence CO_2 uptake, giving SBET a more substantial impact. Second, ML models like LSSVM-GO weigh input features based on how each feature's variability aligns with the target variable—in this case, uptake—across the dataset. Even with high correlation, SBET might demonstrate a stronger alignment with uptake in certain conditions, making the model more sensitive to changes in SBET. Additionally, feature importance techniques, such as SHAP values, reveal complex, nonlinear interactions between SBET and other variables, which can amplify its influence on the predictions generated.

5.5. *Partial dependent plot (PDP) analysis*

PDP offers insights into the factors driving ML/HML model predictions by examining how CO_2-uptake predictions are influenced by adjusting one feature while keeping another constant, which reveals interactions between input features.

Fig. 16 shows PDPs for the LSSVM-GO model predicting CO_2 uptake. The effects of changing input pairs such as T-VT, MC-VT, and MC-T generally follow a linear pattern, depicted as a smooth plane on the plot. However, when other feature pairs are varied, the model output exhibits non-linear and complex changes, indicating deeper complexities in the relationships among input variables and CO_2-uptake. This underscores the necessity for HML models to accurately predict the behavior of MOFs.

6. Limitations and recommendations for future study

While the models developed CO_2-uptake of MOF in this study were configured to achieve their optimal results, some limitations are apparent:

- The dataset size (475 data points) remains relatively small, which may affect the robustness and generalizability of the developed models.

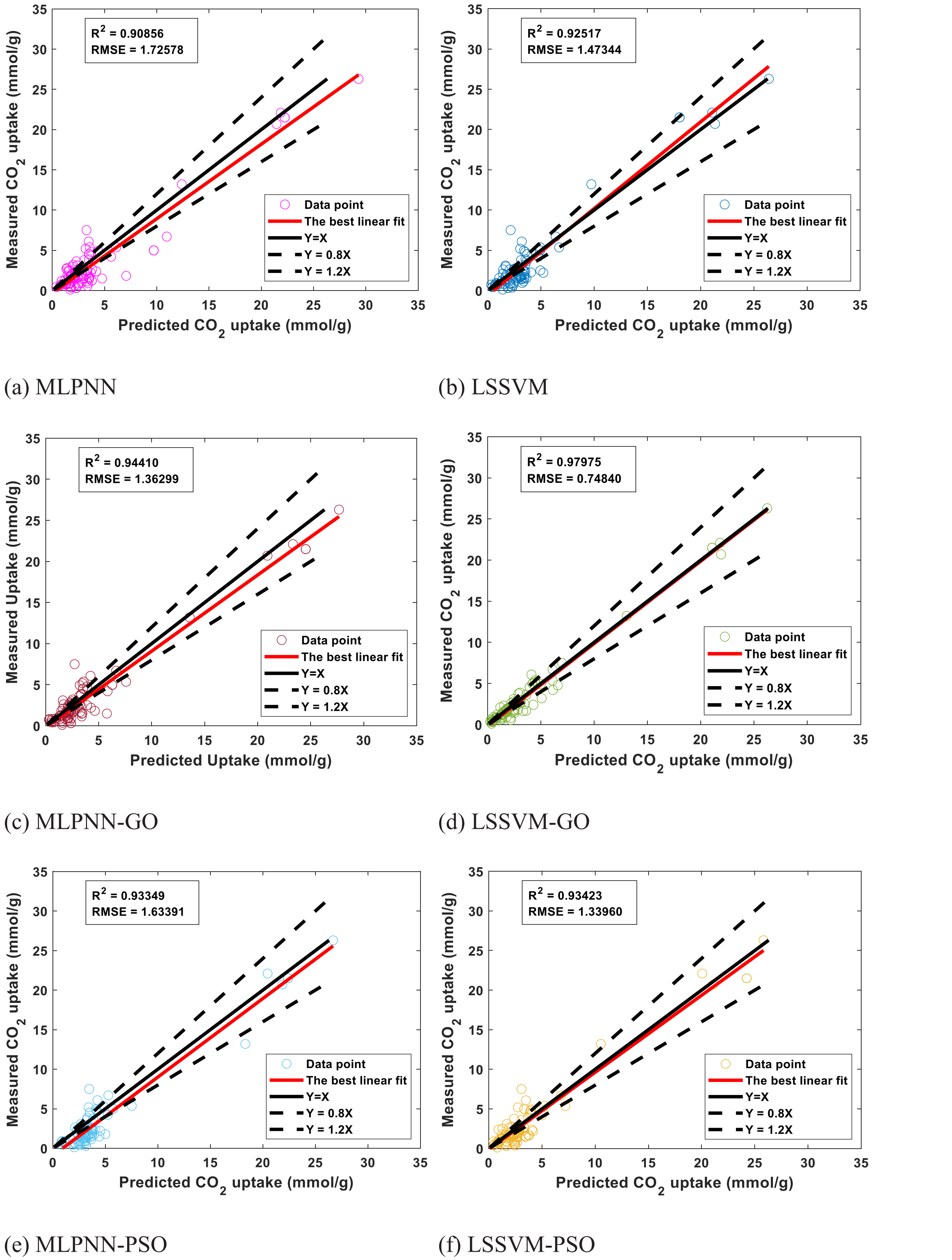


Fig. 11. Cross plots comparing measured and predicted CO_2 uptake utilizing (a) MLPNN, (b) LSSVM, (c) MLPNN-GO, (d) LSSVM-GO, (e) MLPNN-PSO, and (d) LSSVM-PSO trained models applied to the testing data.

Table 5
Comparison of prediction performance for simple and hybrid models applied to the CO_2 uptake testing data.

Type	Models	ARE	RMSE (mmol/g)	RRMSE	R^2	PI	a20-index
Simple	LSSVM	0.6772	1.4734	0.4139	0.9252	0.2110	0.3294
	MLPNN	0.8750	1.7258	0.4848	0.9086	0.2482	0.2824
Hybrid	**LSSVM-GO**	**0.1984**	**0.7484**	**0.2102**	**0.9798**	**0.1056**	**0.6118**
	MLPNN-GO	0.6764	1.3630	0.3829	0.9441	0.1942	0.3294
	LSSVM-PSO	0.5151	1.3396	0.3763	0.9342	0.1914	0.4118
	MLPNN-PSO	1.2019	1.6339	0.4590	0.9335	0.2334	0.2706

Table 6
The OFI values obtained for the developed models based on the results of the training and testing stages.

Model	Simple		Hybrid			
	LSSVM	MLPNN	LSSVM-GO	LSSVM-PSO	MLPNN-GO	MLPNN-PSO
OFI value	0.1873	0.2184	0.0712	0.1534	0.1487	0.1859

- The dataset contains samples with ten specific metal centers—aluminum, nickel, zinc, zirconium, titanium, copper, cadmium, cobalt, iron, and hafnium—analyzed within a limited temperature and pressure range.
- Pre-processing applied to the compiled dataset is of a qualitative nature.
- The reliance on complex measurements, such as BET surface area, may limit the practicality and scalability of the model for real-world applications.

These constraints suggest that future research should focus on:

- Expanding the dataset with additional data points covering a wider range of conditions.
- Considering samples with various numbers of metal centers at a wide range of temperatures and pressures.
- Future work will focus on investigating alternative or proxy variables, such as theoretical surface area, average pore diameter, or framework topology parameters, and integrating high-throughput computational screening data to improve the model's efficiency and applicability in CO_2 capture applications.
- Deep learning algorithms, which can learn complex data patterns, are generally more effective for predicting larger datasets compared to traditional machine learning models.

7. Conclusion

This study developed predictive models for CO_2 uptake in metal-organic frameworks (MOFs) using input variables S_{BET}, P, T, MC, V_T, and S_L. The algorithms used were LSSVM, MLPNN, and hybrid models incorporating PSO and GO. The dataset consisted of 487 points from literature, reduced to 475 after filtering out 12 incomplete records. PSO determined the optimal ratio for training-to-testing separation. The hybrid algorithms were used to identify the best control parameters for model prediction, applied to the training data, and evaluated with test data. Prediction results were checked for overfitting and prediction errors to find the most accurate model. Additionally, SHAP and PDP graphical tools were used to assess the importance of each feature in the best CO_2 uptake prediction model and to understand the marginal impact of each of the six input variables. The findings of this study are:

- PSO determined that the best training-to-testing data ratio is 82 % for training (390 data points) and 18 % for testing (85 data points). This ratio enhanced the models' prediction performance and generalization capability.
- RBF was selected as the optimal kernel function for LSSVM based on a trial-and-error analysis during model development.
- PSO determined the optimal MLPNN structure, featuring three hidden layers with 9, 17, and 9 neurons, respectively. The ideal learning rate was 0.8267.
- The models developed with hybrid algorithms exhibited lower RMSE values than those developed with simple models.
- The LSSVM-GO algorithm showed the lowest RMSE (0.4652 mmol/g for training and 0.7484 mmol/g for testing), indicating the best generalizability among hybrid algorithms. In contrast, the MLPNN algorithm generated the highest RMSE with the least generalizability.
- The LSSVM-GO model demonstrated the least overfitting, while the MLPNN model exhibited the most pronounced overfitting.

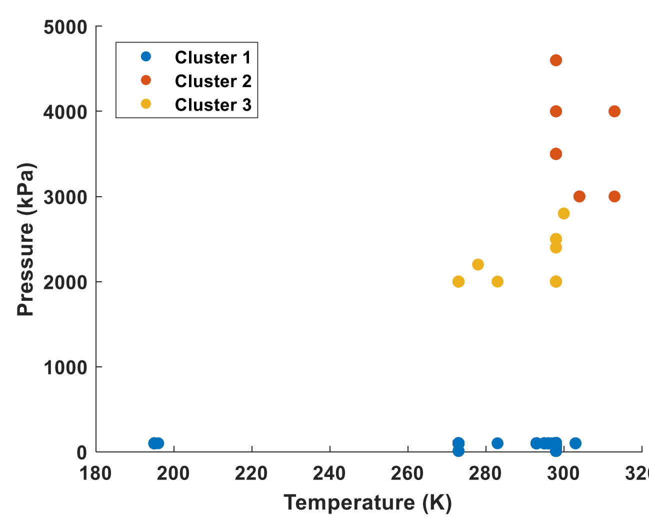


Fig. 12. Scatter plot of temperature versus pressure distinguishing three distinct clusters of data points with similar temperature and pressure conditions.

Table 7
The range of pressure and temperature variations for each cluster, along with the number of data points in each cluster.

Feature	Cluster 1			Cluster 2			Cluster 3		
	Min	Max	Sample count	Min	Max	Sample count	Min	Max	Sample count
Temperature (K)	195	303	429	298	313	23	273	300	23
Pressure (kPa)	10	105		3000	4600		2000	2800	

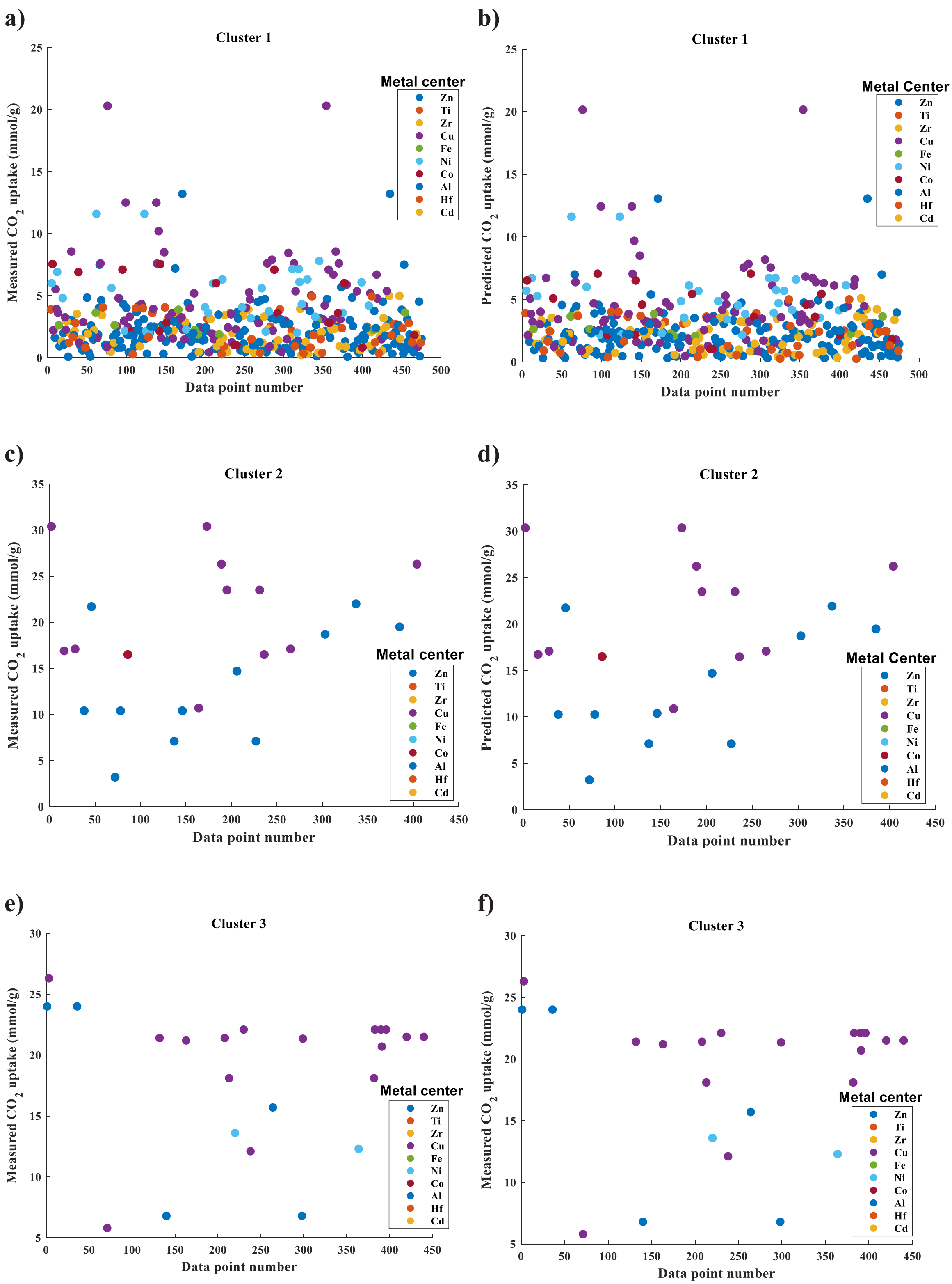


Fig. 13. Comparison of the distribution patterns of measured CO_2 uptake values of MOFs with different metal centers with those values predicted by the LSSVM-GO model across three different clusters of temperature and pressure conditions.

- Combined scoring raked the assessed prediction models: LSSVM-GO (best), followed by MLPNN-GO, LSSVM-PSO, LSSVM, MLPNN-PSO, and MLPNN (worst).
- REC curve comparisons for training and testing stages ranked the models: LSSVM-GO (best), followed by MLPNN-GO, LSSVM-PSO, MLPNN, LSSVM, and MLPNN-PSO (worst).
- Input feature importance analysis for the LSSVM-GO model revealed that S_{BET} exerts the greatest influence and V_T the least impact on the model's output.
- The marginal effects of feature pairs on the LSSVM-GO model's output indicated that interactions between V_T-T, MC-V_T, and MC-T had linear or planar relationships affecting CO_2 uptake prediction.

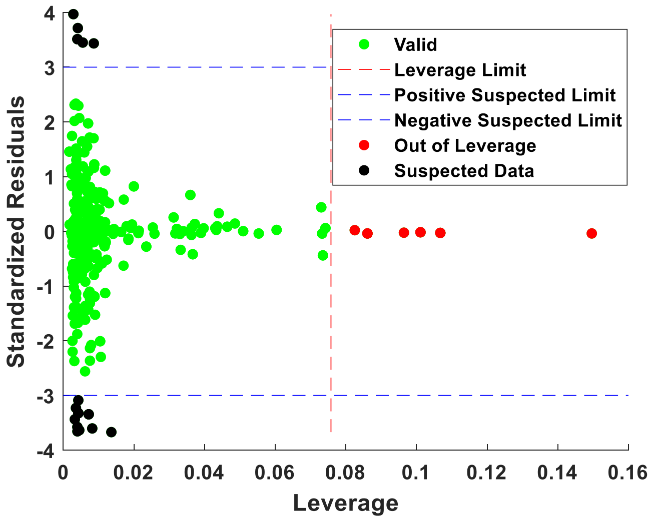


Fig. 14. William's plot displays the CO_2-uptake predictions of the LSSVM-GO model applied to the compiled dataset to identify the domain of applicability and potential outlier data.

Other feature pairs, however, displayed non-linear and complex relationships with CO_2 uptake.

CRediT authorship contribution statement

Promise O. Longe: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Shadfar Davoodi:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Mohammad Mehrad:** Writing – original draft, Validation, Software, Investigation, Formal analysis, Data curation, Conceptualization. **David A. Wood:** Writing – review & editing, Visualization, Validation, Methodology, Formal analysis, Conceptualization.

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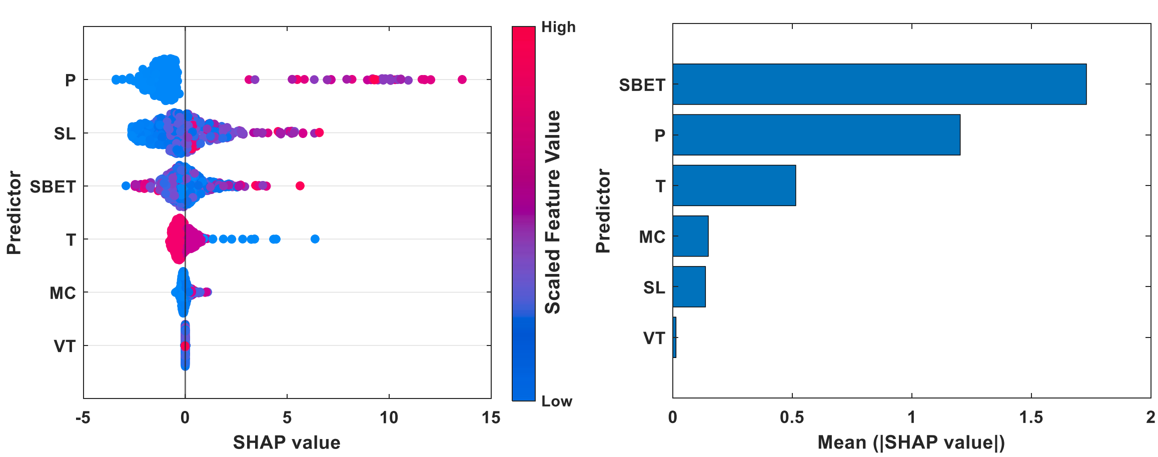
Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. : Evaluation of high uptake values with respect to input variables

The studied dataset contains CO_2 uptake of mainly low values, with relatively few samples displaying higher CO_2 uptake levels. Identifying the conditions associated with high CO_2 uptake can help guide future experiments to replicate these conditions and achieve higher CO_2 uptake values. Therefore, data records of testing samples were divided into two groups: Group 1 for samples with CO_2 uptake values below 15 mmol/g and Group 2 for samples with CO_2 uptake values above 15 mmol/g. Figure A1 presents a box plot comparing the designated groups of CO_2 uptake values with respect to the magnitude of the input features. The results indicate that the samples in Group 2 (CO_2 uptake $>$ 15 mmol/g) generally are associated with higher values for SBET (surface area), SL (adsorption capacity), VT (pore volume), and pressure. These relationships therefore indicate that high CO_2 uptake is associated with the samples with increased surface area, larger pore volume and adsorption capacity, and elevated pressure conditions.



(a) (b)

Fig. 15. Visualizing the influence of each input feature on CO_2-uptake predictions with SHAP values for the LSSVM-GO model: (a) detailed feature impact plot and (b) summary plot of feature importance.

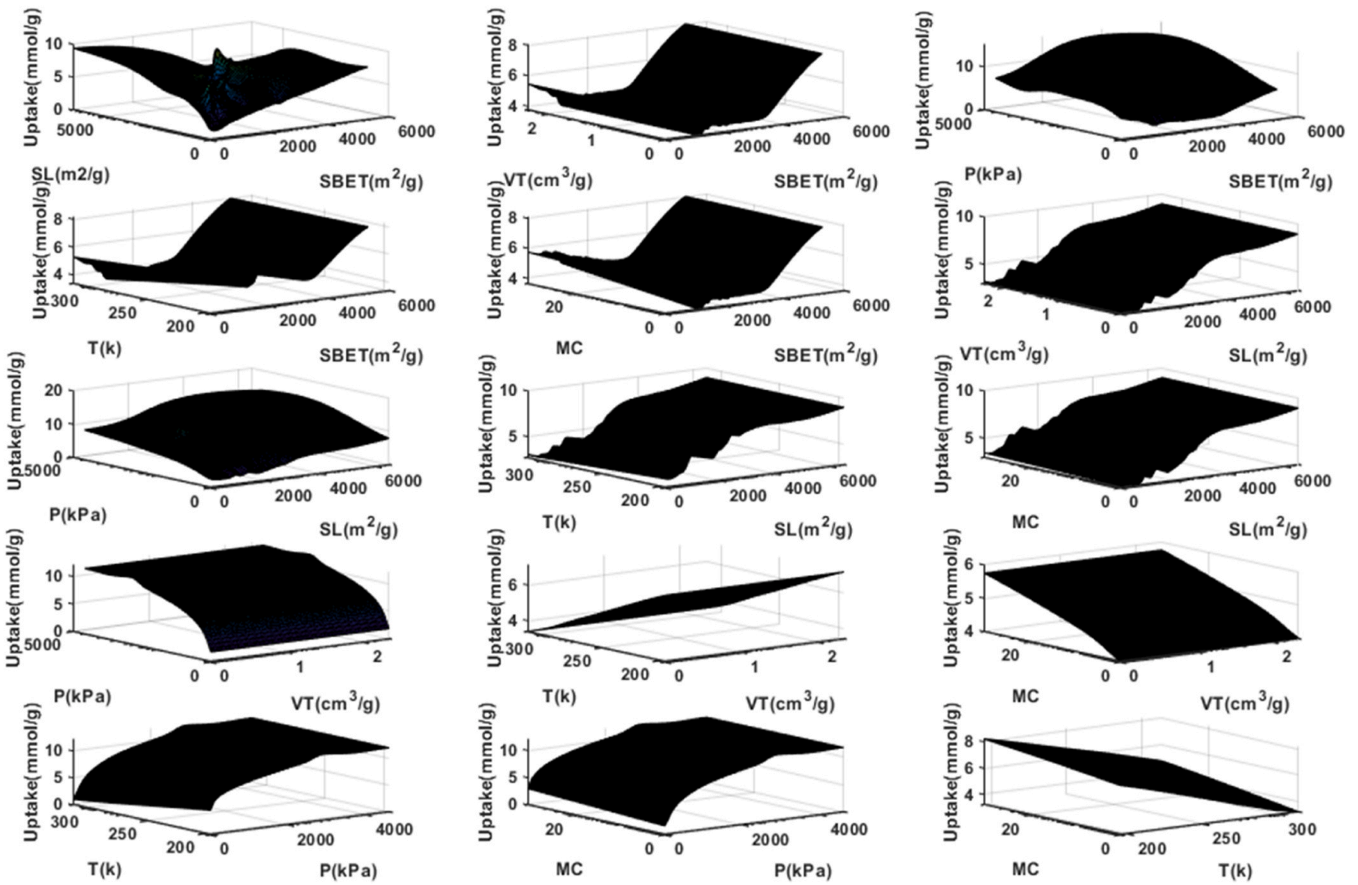


Fig. 16. Evaluation of the effect of pairs of input parameters on the CO_2-uptake predictions of the LSSVM-GO model with the PDP tool.

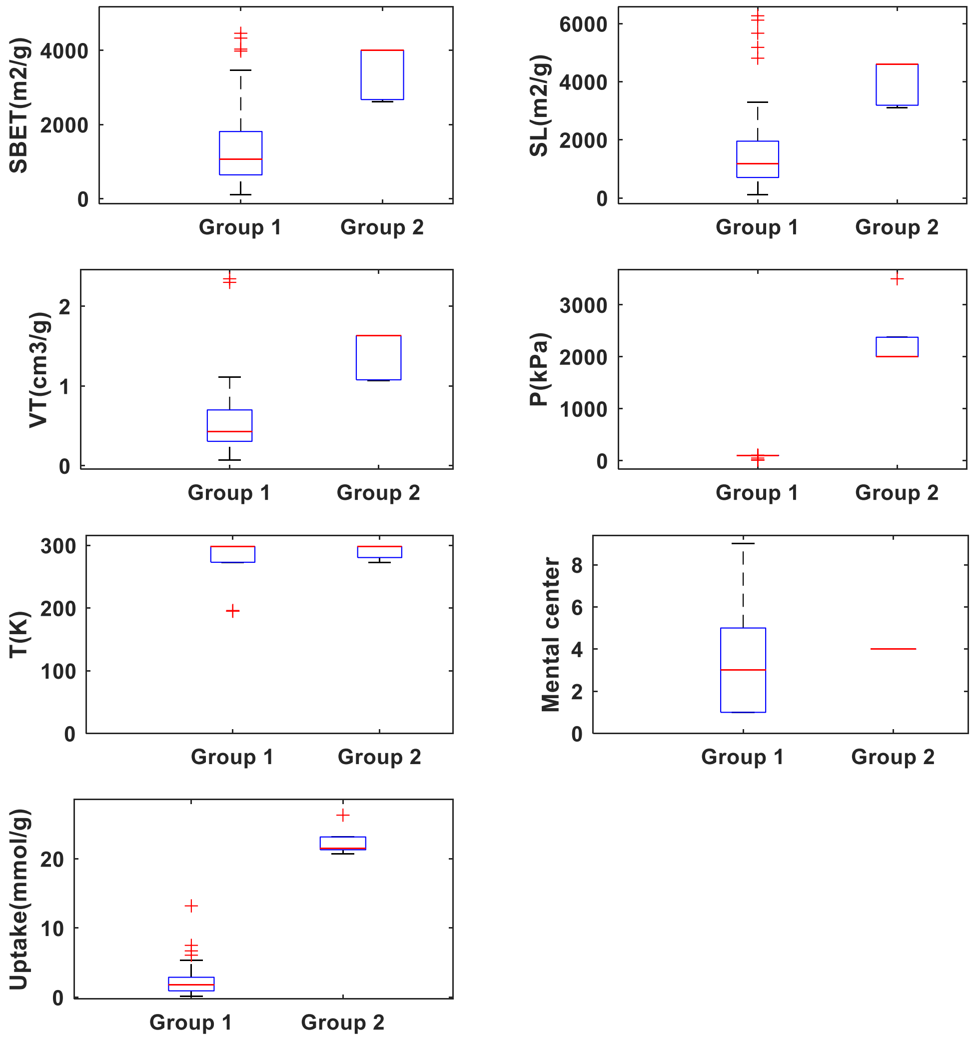


Fig. A1. Box plot comparing two groups of data samples with different CO_2 uptake (Group 1: CO_2 uptake < 15 mmol/g, Group 2: CO_2 uptake > 15 mmol/g) with respect to independent variables' magnitude.

Appendix B. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jallcom.2024.177890.

Data availability

Data will be made available on request.

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